XENON INTERPRETABILITY

Synthetic Market Phase 13

Portfolio-ladder geometry on the same 4-asset market frame used in Phases 9-12. This phase asks whether a second context family deforms the shared market coordinates in the same way as risk, or whether portfolio context behaves more locally and asset-relatively.

DATE	BASE DATA	STATES	OBJECT
22 March 2026	Phase 13 portfolio ladder	row_mean L4, row_eos L14	adjacent portfolio-step transforms

MAIN READ

Phase 13 shows that portfolio context is not just another copy of the risk story. The shared 4-asset market frame still survives almost intact early, but later portfolio effects are best described as local reallocations around that frame rather than as one strong end-to-end global warp.

Early `row_mean @ L4` coordinate transfer stays above `0.989` R^2 for both latent axes across `portfolio_1..portfolio_5`. Later `row_eos @ L14` states still realign toward portfolio-adjusted score geometry, but the transform structure is middle-heavy: `portfolio_1 -> portfolio_2`, `portfolio_2 -> portfolio_3`, and `portfolio_3 -> portfolio_4` admit strong local fits, while `market_only -> portfolio_1` and `market_only -> portfolio_5` remain much weaker.

EARLY FRAME FLOOR	LATE STRONGEST LOCAL FIT	LATE END-TO-END	NONTRIVIAL COMPOSE
0.98921	**0.94023**	**0.49596**	**0.99961**
`latent_x`, `market→portfolio_4`	diagonal on `P2→P3`	identity on `M→P5`	orthogonal matrix cosine

Why Phase 13 Exists

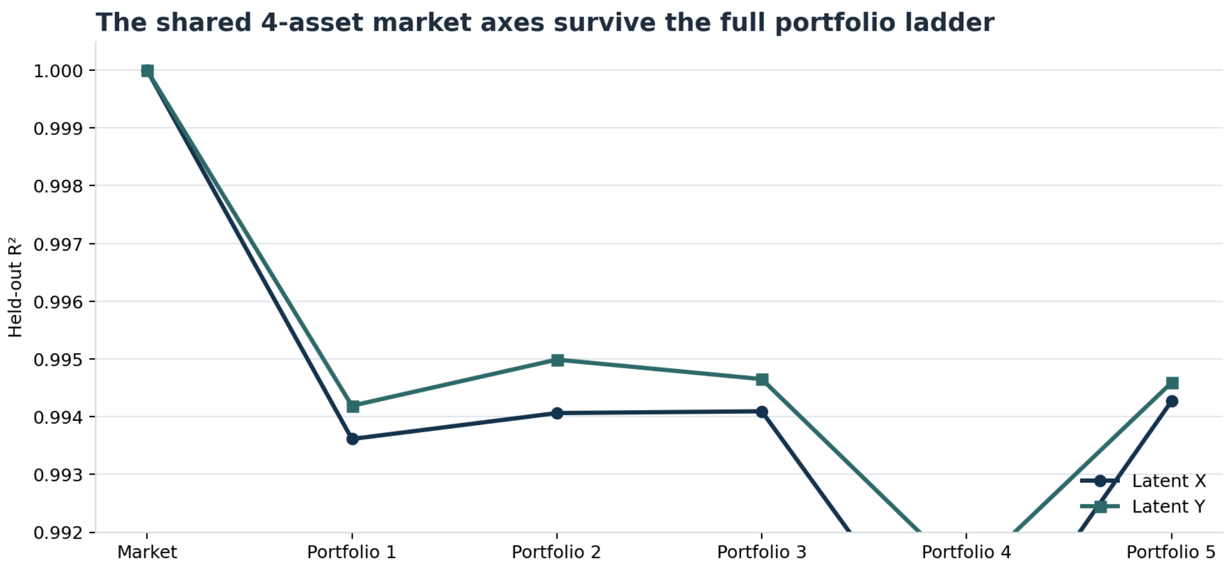
Phase 12 established a strong risk result: the same 4-asset market geometry survives the DX-native risk ladder, and the late end-to-end ladder still composes most coherently under near-rigid maps. The next question was whether that was just a “settings in general” story or something specific to risk.

Portfolio is a useful second family because it is qualitatively different. In the synthetic prompts:

- free ETH decreases across the ladder
- one asset is already held
- the prompt increasingly warns against concentration

That should create an asset-relative overlay. Unlike risk, this context is not supposed to apply the same penalty to every asset.

The Shared Market Frame Still Survives the Full Ladder



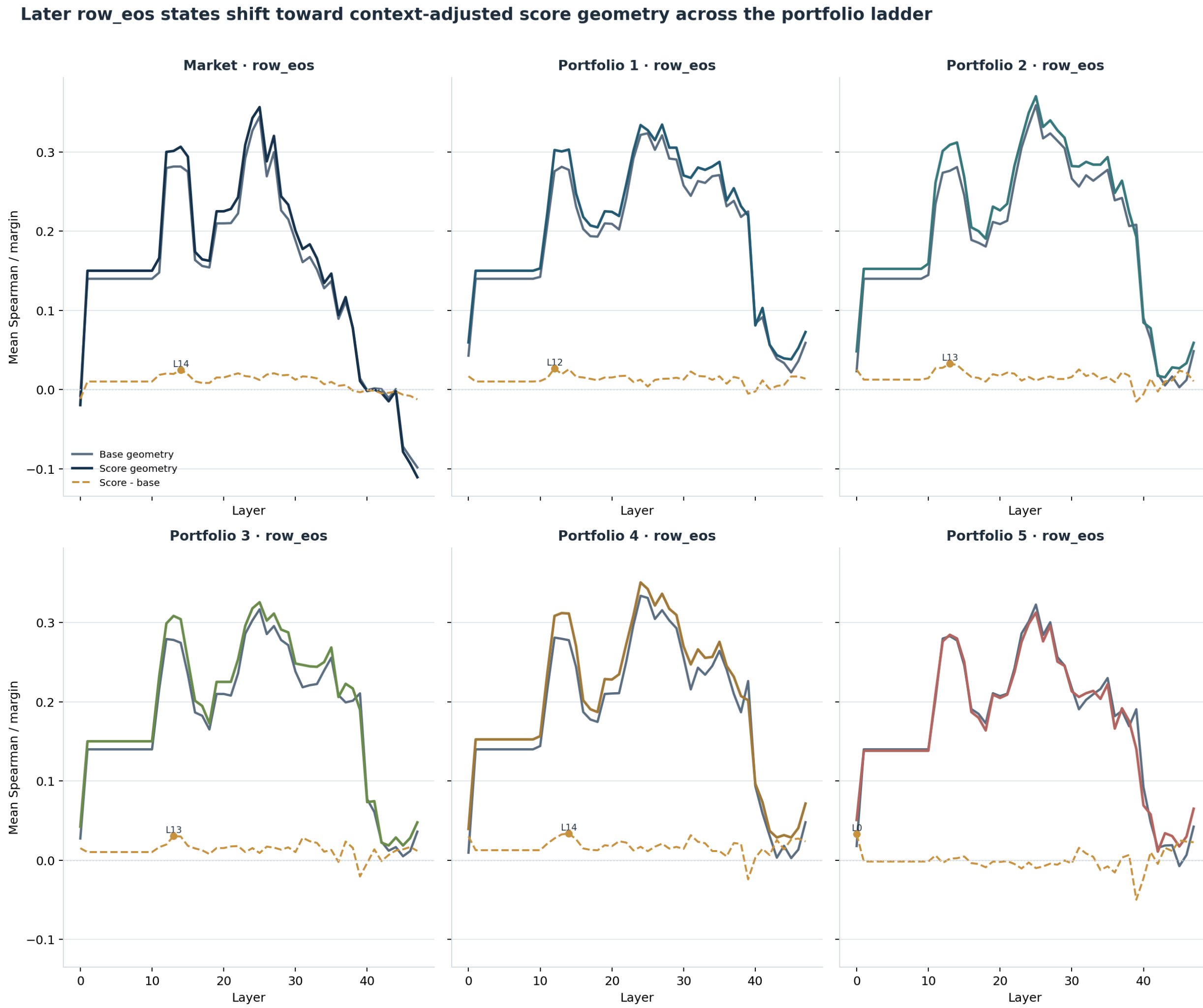
Held-out coordinate transfer from `market_only` into each portfolio context. Both latent axes stay near ceiling, with the best readout always in early `row_mean`.

This is the conservative base result, and it is strong:

- `market_only -> portfolio_1`: `0.9936 / 0.9942` for `latent_x / latent_y`
- `market_only -> portfolio_2`: `0.9941 / 0.9950`
- `market_only -> portfolio_3`: `0.9941 / 0.9947`
- `market_only -> portfolio_4`: `0.9892 / 0.9912`
- `market_only -> portfolio_5`: `0.9943 / 0.9946`

So portfolio context does *not* erase the model's underlying market coordinates. The same 4-asset frame is still present and linearly recoverable very early.

Late States Still Realign Toward Portfolio-Adjusted Score Geometry



Realignment margin between recovered activation-space geometry and portfolio-adjusted score geometry for each context.

As in the risk ladder, the later state moves toward the context-adjusted score geometry:

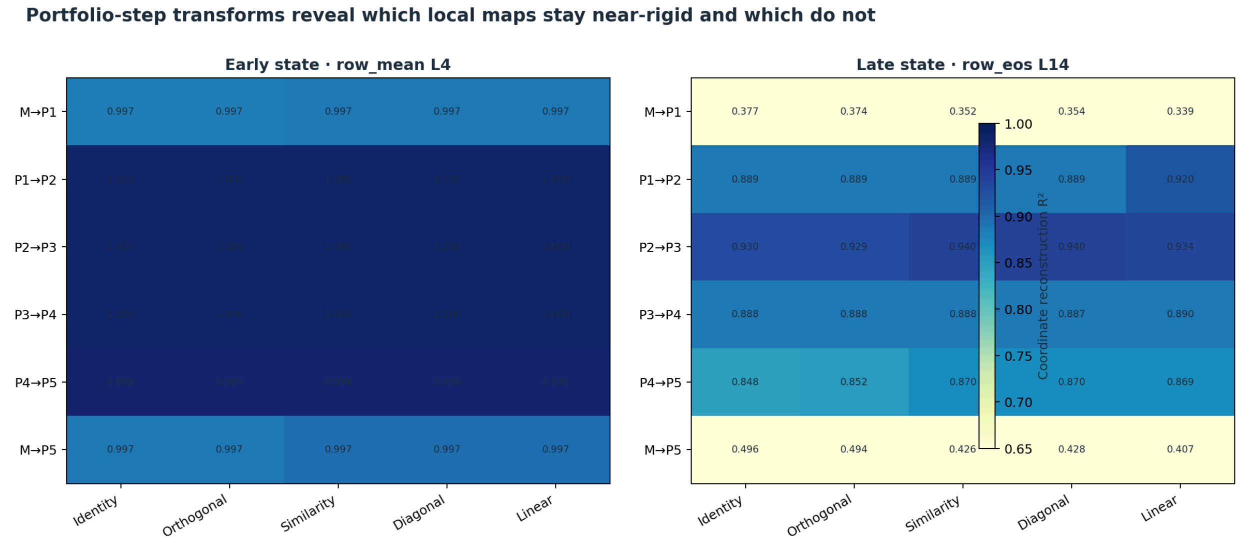
- `portfolio_1`: margin `0.0268` at `row_eos @ L12`
- `portfolio_2`: margin `0.0327` at `row_eos @ L13`
- `portfolio_3`: margin `0.0304` at `row_eos @ L13`
- `portfolio_4`: margin `0.0337` at `row_eos @ L14`

That is a stable late-depth pattern. The one exception is `portfolio_5`, which numerically peaks at `layer 0` but with low NN accuracy (`0.0506`). That looks unstable and should not be treated as meaningful late structure.

The useful reading is therefore:

- early `row_mean` preserves the shared market frame
- later `row_eos` moves that frame toward portfolio-adjusted scores

Early Portfolio Transforms Are Nearly Trivial



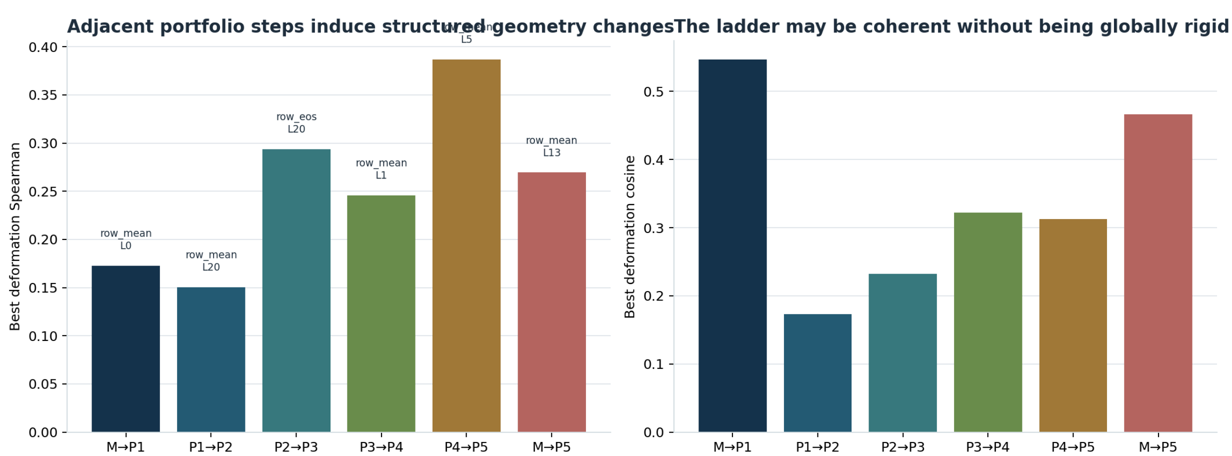
Coordinate reconstruction `R²` for transform families fit on each adjacent portfolio step. Left: early `row_mean @ L4`. Right: late `row_eos @ L14`.

The early panel is almost flat in the same way that the early risk panel was. All adjacent-step fits are already near ceiling:

- `market_only -> portfolio_1`: similarity best, `R² = 0.9969`
- `portfolio_1 -> portfolio_2`: linear best, `R² = 0.9996`
- `portfolio_2 -> portfolio_3`: linear best, `R² = 0.9996`
- `portfolio_3 -> portfolio_4`: similarity best, `R² = 0.9996`
- `portfolio_4 -> portfolio_5`: similarity best, `R² = 0.9994`
- `market_only -> portfolio_5`: similarity best, `R² = 0.9970`

This is the same structural point as before: context insertion is not destroying the market coordinates in the early state.

The Late Ladder Is Local, Mixed, and Middle-Heavy



Best deformation margins for adjacent and end-to-end portfolio comparisons. Higher margins indicate stronger evidence that the geometry is shifting toward the context-adjusted arrangement.

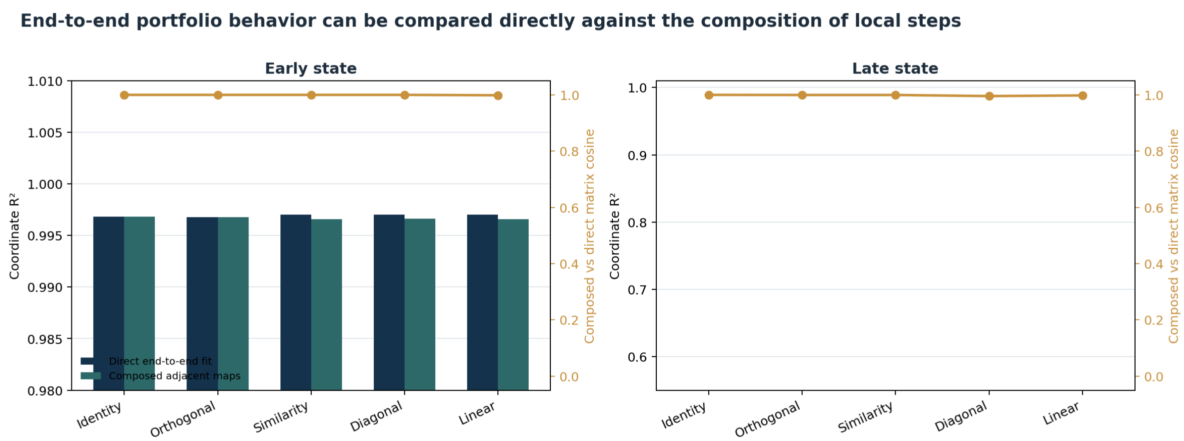
The late transform structure is where portfolio diverges from risk. At `row_eos @ L14`, the strongest local fits sit in the middle of the ladder:

- `market_only -> portfolio_1`: identity best, $R^2 = 0.377$
- `portfolio_1 -> portfolio_2`: linear best, $R^2 = 0.920$
- `portfolio_2 -> portfolio_3`: diagonal best, $R^2 = 0.940$
- `portfolio_3 -> portfolio_4`: linear best, $R^2 = 0.890$
- `portfolio_4 -> portfolio_5`: similarity best, $R^2 = 0.870$
- `market_only -> portfolio_5`: identity best, $R^2 = 0.496$

That pattern matters. The ladder has clean local structure once the portfolio overlay is active, but entering the ladder and the end-to-end `market_only -> portfolio_5` jump do not admit one comparably strong global map.

This is exactly what we would expect if portfolio context mostly changes *relative allocation pressure* inside a preserved market frame. The overlay is not "global caution." It is: "you already hold this one, you have less free ETH, and concentration is now more expensive."

Composition Keeps Orientation Coherent, Not a Strong Global Warp



For `market_only -> portfolio_5`, compare direct end-to-end fits against the composition of all adjacent portfolio-step maps. Gold marks matrix cosine between the composed and direct transforms.

The composition plot needs one careful note: `identity` always composes perfectly by construction, so it is not the informative family here. The more useful comparisons are the nontrivial ones:

- `orthogonal`
  - direct $R^2 = 0.494$
  - composed $R^2 = 0.489$
  - matrix cosine `0.99961`
- `linear`
  - direct $R^2 = 0.407$
  - composed $R^2 = 0.293$
  - matrix cosine `0.99797`

So the end-to-end orientation is still highly coherent, but the portfolio ladder does *not* behave like one strong late global warp. Relative to the risk ladder, this is a more local and more heterogeneous deformation story.

Interpretation

SUPPORTED NOW

The model still carries a shared multi-asset market frame through the full portfolio ladder. Later states then apply context-sensitive local reallocations inside that frame, especially in the middle portfolio steps.

NOT SUPPORTED

A single strong end-to-end portfolio warp is not supported. The `portfolio_5` layer-0 realignment anomaly is also not strong enough to treat as a genuine late-stage effect.

The cleanest synthesis after Phase 13 is:

- risk and portfolio both preserve a shared market frame early
- risk looks globally more coherent as a ladder
- portfolio acts more like an asset-relative redistribution overlay

That is a genuinely interesting result. It says the model is not applying one generic "settings transform" to the market. Different context families appear to deform the same underlying coordinates in different ways.

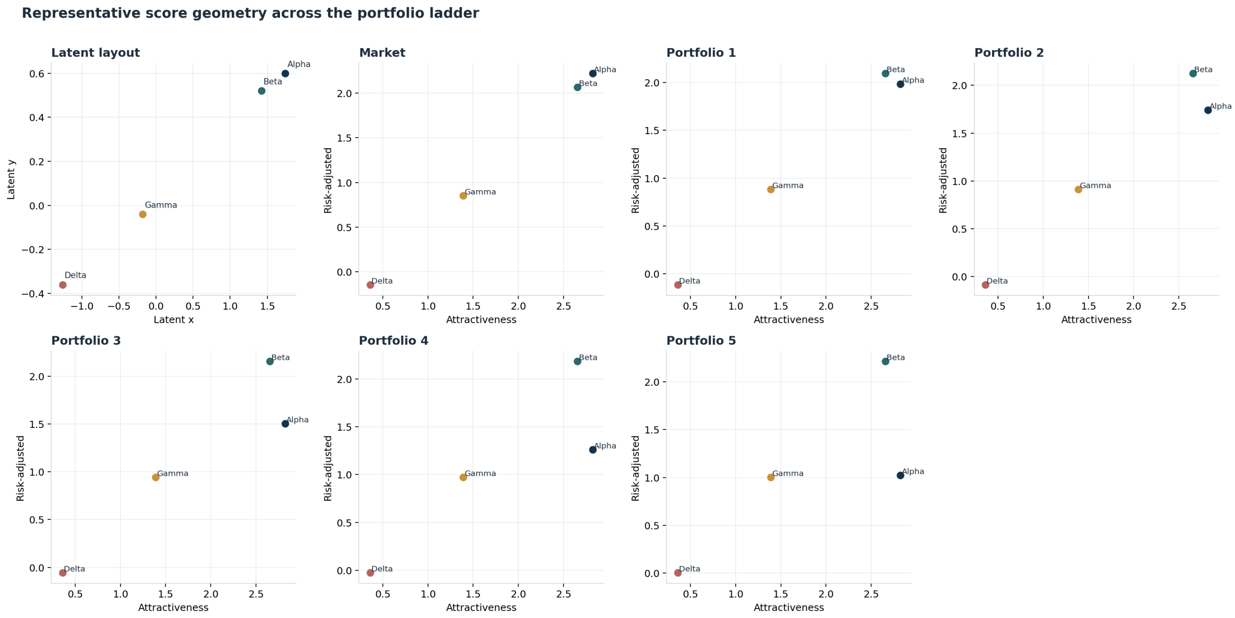
What To Do Next

The next step should keep the same 4-asset geometry object and test one more context family with clearly different semantics:

1. `affordance` / actionability overlays
2. `strategy override` overlays

Then bring the same coordinate-and-deformation lens back to a small real-DX matched-variant set. The goal is no longer "does a latent space exist?" The goal is to test whether real contexts also look like structured deformations of a shared market frame.

Appendix A: Representative Portfolio Geometry



Representative `top_pair_cluster` market from the portfolio ladder. The figure shows the same latent base market with portfolio-adjusted score geometries layered on top of it.

This figure anchors what “portfolio deformation” means in the report. The transform analyses above are not abstract regression exercises. They are attempts to recover how the same underlying 4-asset market is reweighted once an existing position, shrinking free ETH, and anti-concentration pressure are added.

Appendix B: Glossary

Term	Meaning
`shared market frame`	The recovered 2D coordinate system that organizes all four assets before context-specific reweighting.
`coordinate transfer`	How well a probe trained in one context can recover the same latent coordinate in another context. High R^2 means the base frame survived.
`realignment margin`	How much closer the recovered geometry is to the context-adjusted score geometry than to the raw latent layout. Positive margins mean the context is actively reweighting the market.
`identity`	No transform at all. If identity already works, the two contexts share almost the same geometry in the decoded frame.
`orthogonal`	A pure rotation or reflection. Distances and angles are preserved; only orientation changes.
`similarity`	Uniform scaling plus rotation. This captures isotropic compression or expansion.

Term	Meaning
`diagonal`	Axis-aligned rescaling in the shared coordinate frame. This captures simple non-uniform compression without rotation.
`linear`	Any 2×2 linear map. This is the most flexible family here and can absorb rotation, anisotropic scaling, and shear-like behavior.
`matrix cosine`	Cosine similarity between two flattened transform matrices. Here it is used to compare direct end-to-end maps against the composition of adjacent ladder steps.